

Zhi (Leo) Wang

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Education

University of Maryland, College Park

PH.D IN COMPUTER SCIENCE DEPARTMENT

Advisor: **Prof. Yiannis Aloimonos**

Aug. 2025 - Present

College Park, USA

Tsinghua University

B.ENG. IN ELECTRONIC ENGINEERING

Advisor: **Prof. Milin Zhang** and **Prof. Jianyu Chen**

Sep. 2020 - Jul. 2025

Beijing, China

University of Illinois Urbana-Champaign

VISITING STUDENT (ON-SITE) IN COMPUTER SCIENCE DEPARTMENT

Advisor: **Prof. Wenzhen Yuan**

Mar. 2024 - Nov. 2024

Champaign, USA

Research Interests

I work on **efficient robot learning from human data**. My goal is to let robots learn from **multimodal information**—like **human videos and haptics**—to achieve **fine-grained** and **general-purpose robotic manipulation**.

Publications

[C4] HumanEgo: Zero-Shot Robot Learning from Minutes of Human Egocentric Videos

Zhi Wang, Botao He, Kelin Yu, Seungjae Lee, Ruohan Gao, Fulong Huang, Yiannis Aloimonos

IEEE International Conference on Robotics and Automation (ICRA), 2027, Under Review

Best Paper Award, [IROS 2026 WORLDS Workshop](#)

Best Demo Paper Award, [IROS 2026 ScaleInfra Workshop](#)

Featured: [\[Saturday Robotics\]](#), [\[UMD Today\]](#)

[\[Website\]](#), [\[Paper\]](#), [\[arXiv\]](#), [\[Video\]](#), [\[Code\]](#) (★ 563), [\[Dataset\]](#)

[C3] ForceBand: Learning Forceful Manipulation with sEMG

Botao He, **Zhi Wang**, Linna Kuang, Ishaan Ghosh, Jitendra Malik, Cornelia Fermüller, Tingfan Wu, Jiayuan Mao, Ruoshi Liu, Haozhi Qi, Yiannis Aloimonos

Conference on Robot Learning (CoRL), 2026

[\[Website\]](#), [\[Paper\]](#), [\[arXiv\]](#), [\[Video\]](#), [\[Code\]](#) (★ 88)

[C2] DoorBot: Closed-Loop Task Planning and Manipulation for Door Opening in the Wild with Haptic Feedback

Zhi Wang^{*}, Yuchen Mo^{*}, Shengmiao Jin, Wenzhen Yuan

IEEE International Conference on Robotics and Automation (ICRA), 2025

[\[Paper\]](#), [\[Video\]](#), [\[Code\]](#) (★ 57), [\[Website\]](#), [\[Dataset\]](#), [\[UIUC Summary\]](#), [\[UIUC Freshmen Welcome Video\]](#)

[C1] KOSMOS-E: Learning to Follow Instruction for Robotic Grasping

Zhi Wang^{*}, Xun Wu^{*}, Shaohan Huang, Li Dong, Wenhui Wang, Shuming Ma, Furu Wei

IEEE International Conference on Intelligent Robots and System (IROS), 2024, Oral

[\[Paper\]](#), [\[Video\]](#), [\[Code\]](#), [\[Website\]](#), [\[Dataset\]](#)

Research Experience

University of Maryland, College Park (UMD)

Aug. 2025 - Present

PERCEPTION AND ROBOTICS GROUP, ADVISED BY PROF. YIANNIS ALOIMONOS,

College Park, USA

WORKING WITH PROF. FURONG HUANG AND PROF. RUOHAN GAO

- **Research Topic: Robot Learning from Human Data**

- Proposed **HumanEgo**, a robot-data-free, hardware-agnostic, and data-efficient pipeline that learns robot manipulation policies from **minutes of raw human egocentric videos**.
- Designed a **flow-matching policy with dense auxiliary objectives**, enabling **zero-shot** deployment in any environment without specialized workspace or calibration.
- Collected and released a large-scale **human egocentric video dataset**, and validated the learned policy across diverse manipulation tasks and unseen environments.
- **HumanEgo** has received **550+ GitHub stars**, built a **500-person community**, and reached **110k+ views** on X.

Amazon FAR (Frontier AI & Robotics)

May. 2026 - Present

APPLIED SCIENTIST INTERN AT AMAZON FAR (FRONTIER AI & ROBOTICS), WORKING

Bay Area, USA

WITH PROF. GUANYA SHI, PROF. CARLO SFERRAZZA, DR. ROCKY DUAN, PROF.

PIETER ABBEEL, RUOSHI LIU, HAOZHI QI, JIAYUAN MAO, AND PROF. JITENDRA MALIK

- **Research Topic: Forceful and Generalist Robot Manipulation from Human Data**

- Developed **ForceBand**, a low-cost, wrist-worn **sEMG** band that predicts fingertip forces from forearm muscle activity, enabling **force-enriched human demonstrations** for learning forceful robot manipulation.
- Working on **HumanWAM**: training **world action models** from human egocentric videos.

University of Illinois Urbana-Champaign (UIUC)

Mar. 2024 - Nov. 2024

RESEARCH ASSISTANT AT ROBOTOUCH LAB, ADVISED BY PROF. WENZHEN YUAN

Champaign, USA

- **[C2] Research Topics: Mobile and Bimanual Manipulation for Articulated Objects in the Open World**

- Proposed **DoorBot**, a hierarchical, closed-loop, haptic-aware control framework with unified action representation, enabling a bimanual, mobile robot to explore and open diverse unseen doors in the wild. DoorBot achieves a **90%** success rate across 100 trials on totally **20 unseen doors** in the UIUC Campus.
- Main Contributions: **(1) Primitives Design: Well-designed primitives** split the whole task into the high-level planner and low-level policy to lower the dimensionality. **(2) Grasping-and-Unlocking Model: A novel action representation** maps RGB images to 3D action parameters. **(3) Closed-loop System with Haptic Feedback: Robots learn from mistakes adaptively, correct actions autonomously, sense object attributes from haptics.**

Microsoft Research Asia (MSRA)

Jun. 2023 - Mar. 2024

RESEARCH INTERN AT NLC GROUP, ADVISED BY DR. SHAOHAN HUANG

Beijing, China

- **[C1] Research Topics: Multimodal Learning for Semantically Robotic Grasping**

- Proposed **KOSMOS-E**, a Multimodal Large Language Model (MLLM) combining visual and textual information to enhance capabilities for semantically robotic grasping maneuvers.
- Proposed **INSTRUCT-GRASP**, a large-scale, instruction-following, multimodal, robotic grasping dataset comprising **1.8 million** grasping data, 2 modalities, 8 instruction types, 3 information sources, 3 tasks, and 2 scenes.
- KOSMOS-E achieves an **85.19%** success rate for image-wise evaluation and a **72.63%** success rate for object-wise evaluation on Cornell Grasping Dataset.

Institute for Interdisciplinary Information Sciences (IIIS), Tsinghua

Sep.2021 - Sep. 2022

RESEARCH ASSISTANT AT ISR-LAB, ADVISED BY PROF. JIANYU CHEN

Beijing, China

- **Research Topics: The World's Fastest Humanoid Robot in the Wild [Website] [Media]**

- Designed a versatile **humanoid robot** capable of superior locomotion performance in diverse environments, where I finished the efficient **BLDC-FOC motor driver** design and whole-body power supply.
- As the prototype of RobotEra's **STAR1, the fastest humanoid robot** in the open world at that time, it attracts numerous media reports and commercial interests.

Industry Experience

Amazon FAR (Frontier AI & Robotics)

APPLIED SCIENTIST INTERN

Bay Area, USA

May. 2026 - Present

- Robot learning from human data; built **ForceBand** (sEMG force sensing) and **HumanWAM** (world action models).

ZhenFund

INVESTMENT INTERN (VENTURE CAPITAL)

Beijing, China

Mar. 2025 - Aug. 2025

- Sourcing and due diligence on early-stage **AI and robotics** startups.

Microsoft Research Asia (MSRA)

RESEARCH INTERN

Beijing, China

Jun. 2023 - Mar. 2024

- Multimodal LLMs for robotic grasping; proposed **KOSMOS-E** (IROS 2024, Oral).

EncoSmart Technology (Robotics Startup)

ROBOTICS AND COMPUTER VISION INTERN. [\[CODE\]](#) [\[WEBSITE\]](#)

Beijing, China

Apr. 2023 - Jul. 2023

- Topics: Autonomous Frying and Cooking Robot**
- Developed a crucial vision module for robotic grasping and insertion, helping LAVA, an autonomous cooking and frying robot, achieve **sub-millimeter accuracy**.

Leaderships & Activities

Chair of the Electronic Engineering Hardware Group

30-PERSON TEAM, TSINGHUA UNIVERSITY [\[WEBSITE\]](#)

2021-2023

Tsinghua University

- Organized two major annual, university-wide competitions, engaging over **450 participants**.

Leader of Hardware and Vision Team in Future Robot Club

15-PERSON TEAM, TSINGHUA UNIVERSITY [\[WEBSITE\]](#) [\[GITHUB\]](#)

2021 - 2023

Bordeaux, France

- Led the team of Tinker, a domestic service robot, participating in annual **RoboCup@Home** Competition.

Teaching Experience

Fall 2026 **Graduate Teaching Assistant**, CMSC426: Computer Vision

UMD

Fall 2025 &

Spring **Graduate Teaching Assistant**, CMSC420: Advanced Data Structure

UMD

2026

2022 **Head Teaching Assistant**, 40231212: Intelligent Robots Design and Implementation

Tsinghua University

2021 **Head Teaching Assistant**, 20230292: Project Design and Making of Electronic System

Tsinghua University

2021 **Head Teaching Assistant**, 01550013: Synthetical Practice of Electronics System Design

Tsinghua University

Honors & Awards (selected)

Oct. 2024 **Science and Technology Innovation Scholarship (1 %)**, **Three times: 2021, 2023, 2024**

Tsinghua University

Oct. 2024 **Grand Prize of International Study Scholarship (0.2 %)**, **Only 2 in 1018 People**

Tsinghua University

Jul. 2023 **Top 8 in the world in RoboCup@Home Competition (2 %)**, Domestic Service Robot

Bordeaux, France

Dec. 2021 **Winner Price (5 %)**, The 23rd Electronic Design Competition

Tsinghua University

Apr. 2021 **Third Prize (5 %)**, The 4th Software Design Competition

Tsinghua University

Apr. 2021 **Third Prize (5 %)**, The 4th Artificial Intelligence Challenge

Tsinghua University

Skills

Programming	Python (PyTorch, NumPy, OpenCV), C/C++ , Linux Shell, MATLAB, Verilog, $\text{\LaTeX}$
Software Tools	Git, Anaconda, Docker, ROS1/2 , PyBullet
Hardware System	Mechanics (Solidworks, Blender, Cura), Electronics (STM32, ESP32, Arduino, FPGA, PCB Design)
Robotics	<u>Gelsight Mini</u> , <u>UR5e Robot Arm</u> , <u>RealMan Humanoid Robot</u> , FR5 Robot Arm, Realsense, Kinect